ETHINYLESTRADIOL + LEVONORGESTREL

VTE risk
Check BP & smoking

CHEMICAL NAME Ethinylestradiol: 19-nor-17α-pregna-1,3,5(10)-trien-20-yne-3,17-diol + levonorgestrel: (17α)-13-ethyl-17-hydroxy-18,19-dinorpregn-4-en-20-yn-3-one
CLASS Combined oral contraceptive pill (COC) — oestrogen + progestogen
TRADE NAMES Microgynon 30 · Rigevidon · Nordette · Levlen

THERAPEUTIC USES / INDICATIONS

- Contraception.
- Menstrual disorders: dysmenorrhoea, menorrhagia, irregular cycles, PMS.
- Acne, PCOS symptoms (hirsutism, cycle control).
- Endometriosis symptom control.

ADMINISTRATION

- **PO** one tablet daily at about the same time for 21 days, then 7-day break (or tailored/continuous regimens).
- Start day 1–5 of cycle (immediate protection) or any day + 7 days condoms.
- Missed pill rules: if >24 h late (≥1 missed pill), take latest, use condoms 7 days; may need emergency contraception if in week 1.

DOSAGES

- **Microgynon 30:** ethinylestradiol 30 mcg + levonorgestrel 150 mcg once daily for 21 days, then 7 pill-free days.
- Vomiting within 3 h or severe diarrhoea = missed pill.

PHARMACOKINETICS

- **Onset:** contraceptive after 7 consecutive pills.
- Ethinylestradiol: first-pass metabolism, enterohepatic recirculation; metabolised by CYP3A4.
- Excreted in urine and faeces. **t½:** EE 13–27 h; LNG 24–32 h.

PHARMACODYNAMICS

- Inhibit ovulation (suppress FSH/LH via negative feedback).
- Thicken cervical mucus; thin endometrium.
- Regulates cycle; ↓ androgen activity (↑ SHBG).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, breast tenderness, headache.
- Breakthrough bleeding (first 3 months).
- Mood changes, bloating, ↓ libido.

ADVERSE EFFECTS (SERIOUS)

- Venous thromboembolism (DVT, PE) — ↑ with smoking, obesity, immobility, age >35.
- Arterial thrombosis: MI, stroke (especially migraine with aura, smokers >35).
- Hypertension; cholestatic jaundice; small ↑ breast/cervical cancer risk.
- Liver adenoma (rare).

CONTRAINDICATIONS & CAUTIONS

- Smoker ≥35 yr (≥15/day); BMI ≥35 (relative).
- Migraine with aura; history of VTE, stroke, IHD; thrombophilia.
- BP ≥160/100 (or ≥140/90 relative); breast cancer; severe liver disease.
- <6 weeks postpartum if breastfeeding; major surgery with prolonged immobilisation.

KEY INTERACTIONS

- Enzyme inducers (rifampicin, carbamazepine, phenytoin, topiramate, St John's wort, some HIV drugs) → contraceptive failure.
- **Lamotrigine:** ↓ lamotrigine levels.
- **Ciclosporin:** ↑ levels.

NURSING MANAGEMENT

- Use UKMEC/WHO-MEC eligibility criteria: BP, BMI, smoking, migraine, VTE history.
- Teach missed-pill rules and when extra precautions are needed.
- Annual BP check.
- Stop 4 weeks before major surgery (or use thromboprophylaxis).

PATIENT EDUCATION

- Take at the same time daily; know the missed-pill rules.
- Use condoms with enzyme-inducing drugs and to prevent STIs.
- Seek urgent help for calf pain/swelling, chest pain, breathlessness, severe headache, visual changes (ACHES).
- Stop smoking.

KEY POINTS

- COC — inhibits ovulation.
- VTE/stroke risk; check smoking, BP, migraine with aura.
- 21 days + 7 pill-free days.
- Enzyme inducers cause failure.

SOURCES CHECKED **[675]** Levonorgestrel and Ethinyl Estradiol, oral (US label) **[676]** levonorgestrel and ethinyl estradiol (US label)
Full references p. 432–457